

STUDY OF AC TO DC CONVERTER

Abstract

Modern electrical equipment has a low DC output voltage. The alternating current utility supply must be converted and stepped down to a sufficient DC voltage. Previously, converters were enormous in size and cost a lot of money. The discovery of the switch mode power supply enabled a significant decrease in size and cost. This aim is achieved via increasing power density, which is accomplished by reducing the size of passive/energy storage components such as inductors, capacitors, and transformers.

For better usage of existing electric energy, the complete power system should be run at high efficiency, which is why numerous types of converters are currently used. The research of AC-DC converters is carried out in this thesis, together with necessary simulations, to achieve a greater grasp of the issue. As a result, the primary goal of this research is to thoroughly investigate the working principles and applications of linear power supply in ac to dc converters.

I. INTRODUCTION

At present time there is an increasing demand for efficient systems every time we communicate approximately electricity consumptions and so as to keep up with those needs engineers were coming forward growing efficient conversion strategies and additionally been capable of design circuits with excessive performance. Nevertheless, technology on this discipline remains improving with facing new demanding situations. Converter in power electronics converts power from one type to another by means of various the voltage or frequency. Power conversion is based on kind of the input and output power.

Converter driven programs are broadly used. The electrical power conversion is an exceptional field of electrical engineering. The Power electronics is the utilization of solid-state gadgets to control of electric power and transformation of electric power. It is also restrictive as it excludes rotating machines (electrical motor and generator). Converter may do one or more functions and give an output that varies from the input. The converters are used to increase or decrease the magnitude of the input voltage, invert polarity or produce several output voltages of either the same polarity with the input, different polarity, or mixed polarities.

The converters are used in nonlinear components. For example, the semiconductor switches, and linear reactive components. There are different types of linear components such as the inductors, transformers and capacitors for intermediate energy storage as well as current and voltage filtering. The size, weight and cost of the converter are mainly determined by these components. However, power conversion systems can be classified according to whether the input and output are alternating current (AC) or direct current (DC).

Alternative current (AC) to Direct current (DC) is one of those heavily used conversion in power electronics system. For instance, a laptop battery adapter is the example of the AC to DC conversion. Here, adapter 's input current is AC and output current is DC. The reason behind for using the constant DC voltage is that it is free of ripples or unwanted frequencies. The DC circuit is formed in such a way that the rectification and filtering process will be carried out properly. If the supply of DC voltage is not constant, a laptop cannot be charged properly.

The main feature of the rectifier is it must handle the output current (load). After that, we connect the output of rectifier to the filter. As we know, filter is nothing but a device, which

filters out the ripples in the unidirectional ac voltage and converts it back to pure dc voltage. There are two kinds of filter used in the circuit, inductor and capacitor. Capacitors are the widely used filters. But instead of giving less value if we increase the value of capacitor some circuit, we can get voltage greater than or equal to the dc output voltage (that must be same of transformer output voltage) so that you'll get a pure dc output without any ripples.

Theoretically, it is recommended to use a high voltage capacitor (like 150% of output voltage) to get less ripple and a smooth graph. We try to add a couple of capacitors to get our desired output dc voltage lesser than the transformer output voltage, that is why we will connect a voltage regulator next to the capacitor. We are using 230V/50Hz AC supply which connect to step-down transformer which will step down and give a DC voltage depending on transformer rating that connect rectifier circuit, which converts from AC to pulsating DC after that connect Capacitor (filter).

Afterwards it converts pulsating AC to pure DC. In order to get constant DC, we need to connect regulator IC - LM7812 that maintains the output voltage at a constant value. We are using LM7812 IC, which provides +12 volts regulated power supply. In addition, we try to collect the different changing value for changing the AC voltage and will try to simulate the output waveform. For this purpose, we will use PROTEUS software to get our desired result and waveform.

Background study

The transmission of electric power is in an alternating current (AC) system, with a few exceptions. With the advancement of alternating to direct current converters, they are becoming a recognized and widely used application. As a result, ac dc converter has recently framed a functional research domain. The need for DC power might be supplemental, such as in electronic controls, or critical, such as the DC connection of a motor.

Meanwhile, governing authorities, such as IEC 1000, Std 500, and others, maintain rigorous harmonics guidelines. This is due to the requirement for a strong factor and minimal line current harmonics distortion. To change line frequency from ac to dc, a line frequency diode bridge rectifier is utilized. A large filter capacitor is utilized at the rectifier output to diminish the ripple in the output voltage. In any case, current drawn by this converter is peaky by characteristics for the larger capacitor.

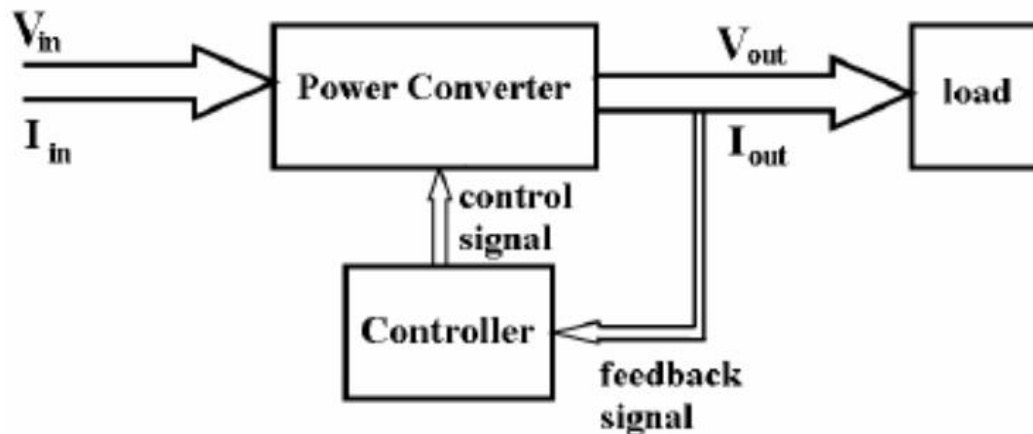
This input current is high in low order harmonics and because of the existence of these harmonics, the aggregate harmonic distortion is high and the input control factor is denied. Issues related with these low power factors also, harmonics, utilities will authorize harmonic standards and rules which will constrain the measure of current distortion permitted into the utility. It is greatly required to accomplishing rectification at near utility power factor and low input current distortion.

II. THEORY & METHODOLOGY

AC-DC converters are electrical circuits that transform alternating current (AC) input into direct current (DC) output. They are used in power electronic applications where the power input a 50Hz or 60Hz sine-wave AC voltage that requires power conversion for a DC output. We utilize control electronic converters in our day-byday life at home, business work places or in a mechanical situation.

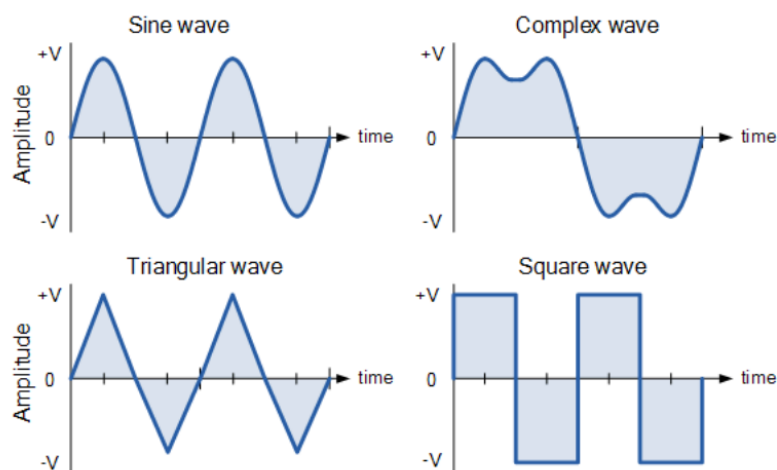
Converters have become an integrated part of industrial electric drives, power supplies and other automobile equipment's due to high power handling capability with higher

efficiency. Power electronic components such as, Silicon Controlled Rectifiers (SCRs) [or Thyristor], Triode for Alternating Current (TRIACs), Insulated Gate Bi-polar Transistor (IGBT) are used for power electronic converters to control and change over electric power. The principle point of the converter is to deliver moulding power regarding a specific application.



Block diagram of a power electronic converter

The block diagram of a power electronic converter shows that it contains an electrical energy source, power electronic circuit, a control circuit and a load circuit. This converter changes one form of electrical energy to another. Commercially available voltages are time varying voltages used in large quantities. Alternating current (AC) is the electric current which periodically reverses the flow of direction. Commercially a variable AC wave forms are **Sinusoidal, Square wave, Complex wave form and Triangular wave.**

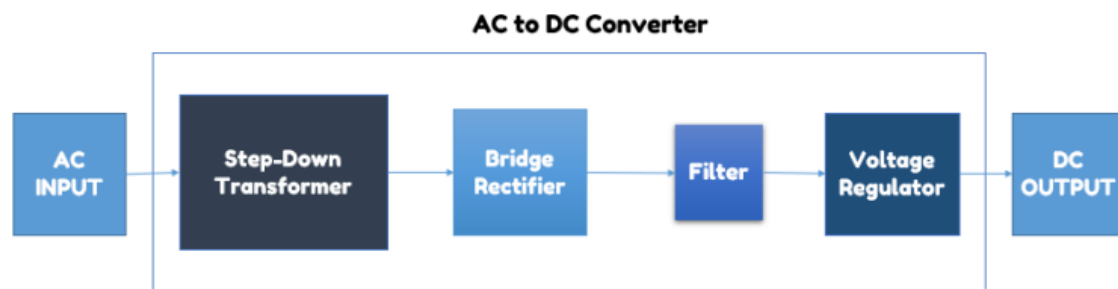


Alternating waveforms.

We are interested in Sinusoidal ac voltage since it is encountered and utilized all over the world. The process of converting AC current to DC current is also known as rectification. We utilize alternating current in our house, but we cannot always use alternating current everywhere. Sometimes we need to convert this to a direct current source. That is why correction is required. The term "rectifier" refers to an AC to DC converter.

The circuit converts an alternating current signal to a direct current signal by allowing the signal to flow in just one direction. Filters are used to smooth down the output even further. Some examples of AC-DC converters are: 1) AC/DC converters (off-line) 3ph-Input/3KW 2) AC/DC converter (off-line) as insert unit 3) AC/DC converters (off-line) with 3 outputs, pen frame 4) AC/DC converters (off-line) as Build-In- Unit 6HE 5) AC/DC converters (rectifier) for Telecom Application

Basic Principles The block diagram of a basic AC-DC converter is given below.



Block diagram of an AC-to-DC converter

To bias all electronic circuits, a DC power source is necessary. Depending on the electronic application, the DC output V_o will typically be in the 3-24V range. The diode is highly beneficial in this case since the properties of a silicon diode are non-linear, which means that current occurs for one voltage polarity but is zero for the opposite polarity.

There are several methods for correction. We'll go through them all afterwards. The simplest powerelectronics circuit used in radios for low-cost power supply is the halfwave rectifier. However, for most power applications, half wave rectification is inadequate. Because the waveform's harmonic richness is so high, filtering the output is extremely challenging. The essential ideas of this rectifier are fairly simple to grasp. Let's look at the fundamentals of a half-wave rectifier.

Half Wave Rectifier

During each positive half cycle of the sine wave, the diode is forward biased as the anode is positive so the current flows through the diode. The diode is in reverse biased during the negative half cycle which blocks the current flow. This way all positive half cycle waveforms are made unidirectional. Using a capacitor parallel to the load the amount of ripple voltage can be reduced and a steady current is produced.

Full Wave Rectification

The full wave rectification uses both positive and negative half cycles of the sine wave. In case of full wave rectification, two diodes are used of which one conducts during the positive half cycle and the other conducts during the negative half cycle. The important point is that the current through the load flows in the same direction thus the output is unidirectional. Turning on a diode depends on the polarity of the cathode voltage and turning it off depends on the device current.

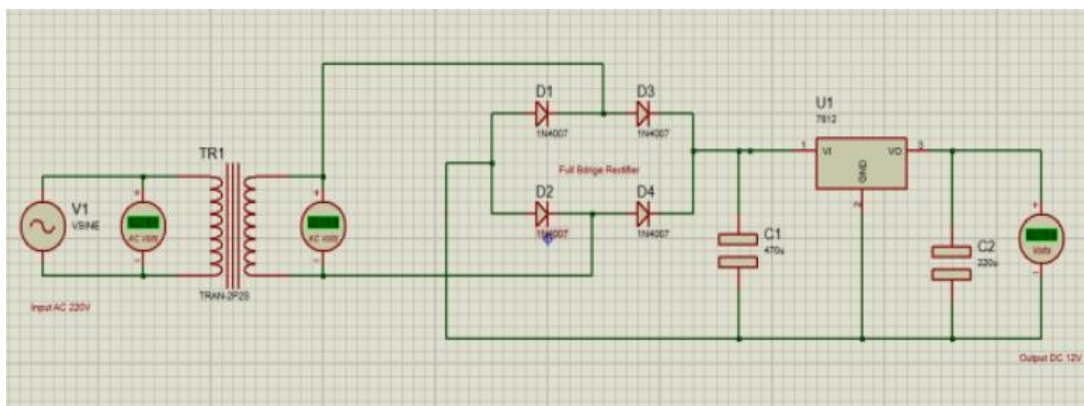
Three terminal devices such as Thyristor, BJT and MOSFET etc. can be used to control the output voltage. The conversion of AC-DC will take place when input is an AC supply and the output is a DC load for example battery charging, DC motor, electrolysis, electromagnets

etc. In designing a rectifier, two important parameters must be specified: the current handling capability and the Peak Inverse Voltage (PIV) the diode must withstand without breakdown. PIV is specified by the manufacturer.

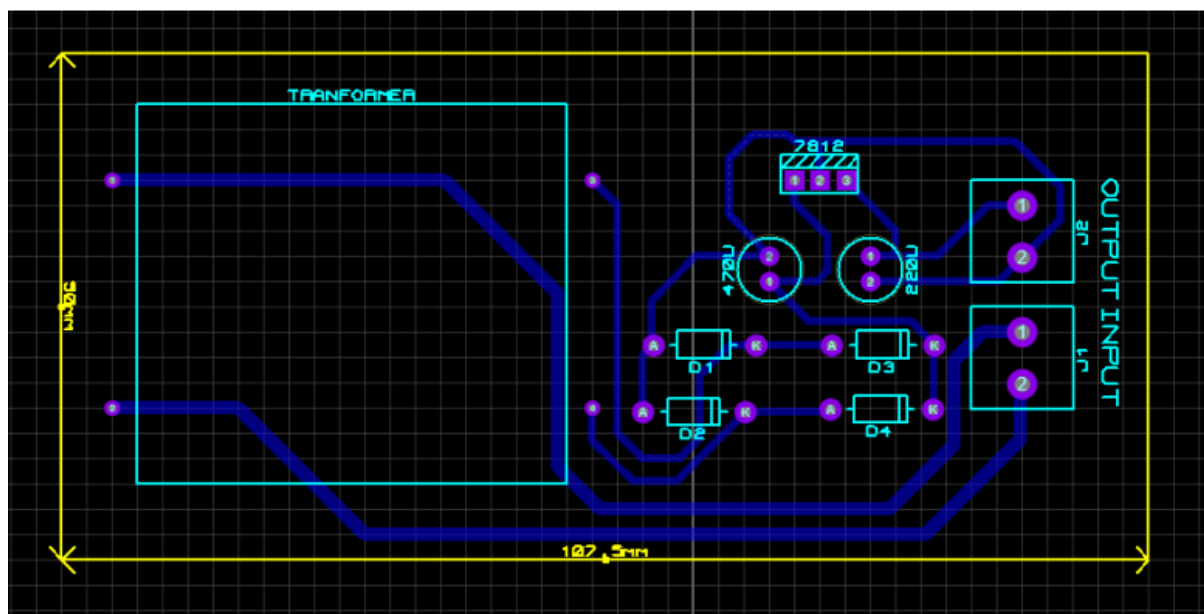
III. LAYOUT DESIGN

Software: Proteus 8 Professional

There is various way of getting AC to DC voltage converter. We can use power electronic converters such as step-down converter, step-up converter, voltage stabilizer, AC to DC converter, DC to DC converter, DC to AC converter. Here, we are doing analysis by using Single Phase Transformer with Full Bridge Rectifier.



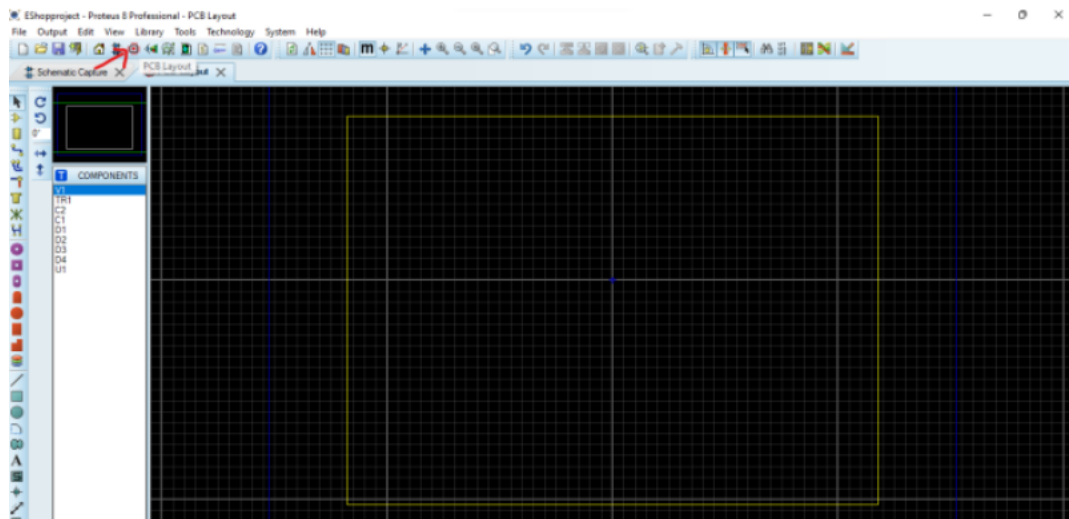
Circuit Diagram of AC voltage as input with output of Full Bridge rectifier.



PCB Layout of the circuit

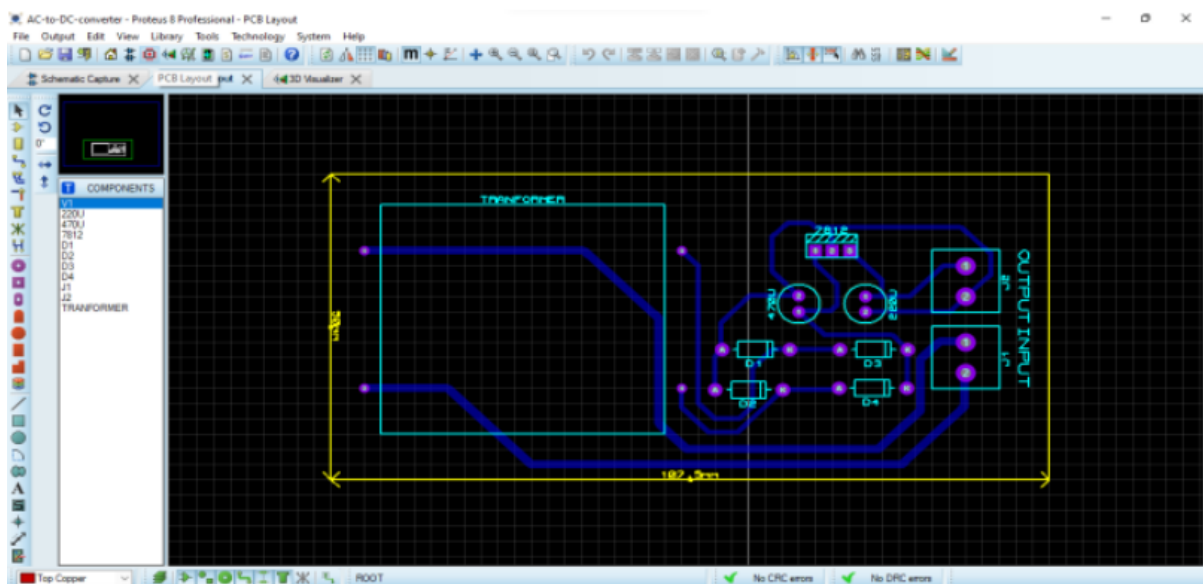
IV. CREATING THE PCB LAYOUT

Open your PCB Layout in the top icon section, and click on the PCB Layout button:



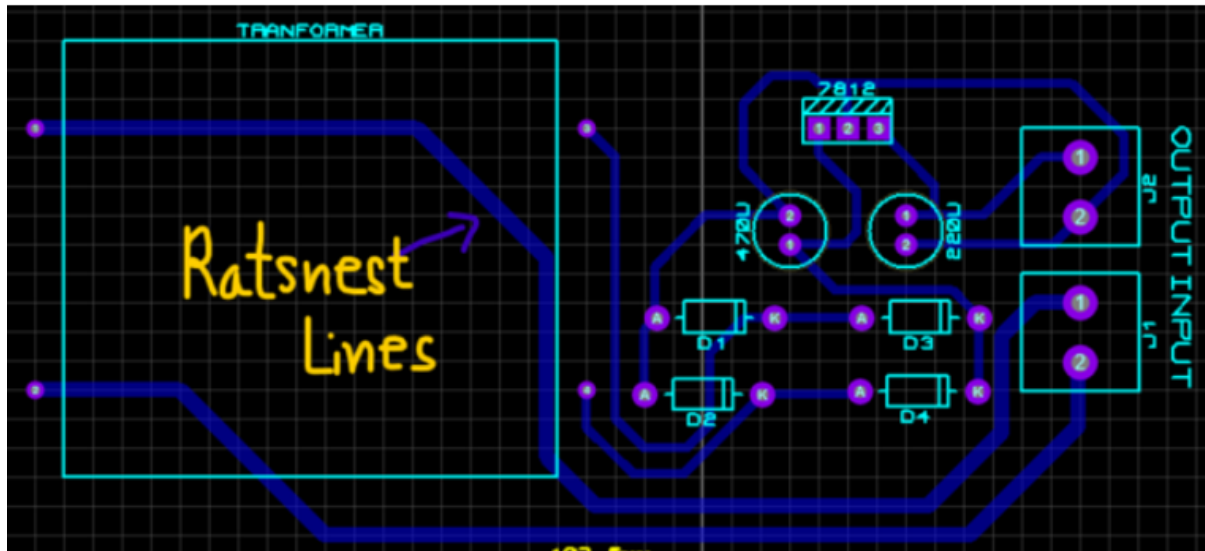
Creating PCB layout interface

The footprints associated with each schematic symbol will be automatically transferred to the PCB editor using auto placer option:



PCB layout Manual Routing

Notice the thin blue lines connecting the components. These are called ratsnest lines. Ratsnest lines are virtual wires that represent the connections between components. They show you where you need to route the traces according to the wiring connections you created your schematic:

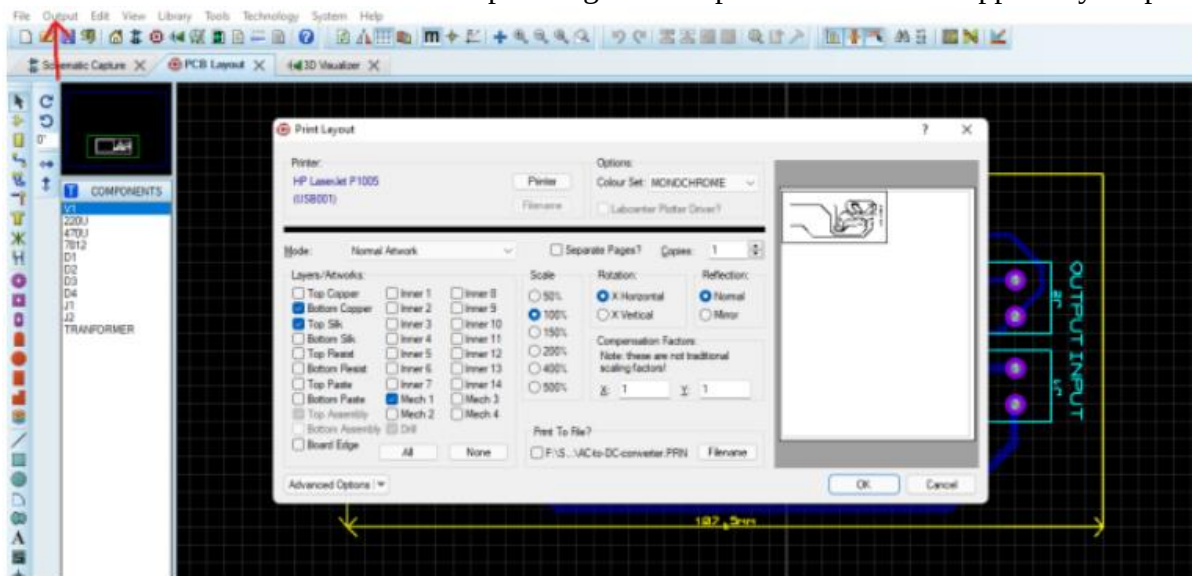


PCB layout Ratsnest Lines

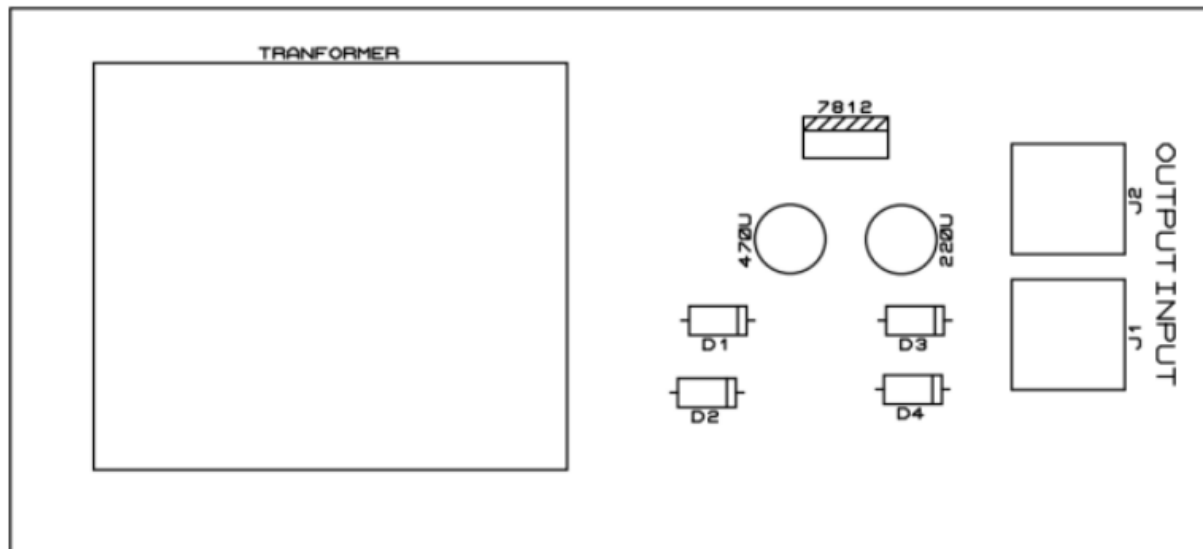
Now we can start arranging the components, keeping in mind the design tips mentioned above. we might want to do some research to find out if there are any special design requirements for our circuit. Some circuits perform better with certain components in specific locations. For example, in an LM7812 amplifier circuit the power supply decoupling capacitors need to be placed close to the chip to reduce noise.

After we've arranged all of the components, it's time to start drawing the traces. Use the ratsnest wires as a rough guide for routing each trace. However, they won't always show the best way to route the traces, so it's a good idea to refer back to your schematic to verify the correct connections. Traces can also be routed automatically using the software's auto-router. For complicated circuits, it's generally better to route traces manually, but try the auto-router on simpler designs and see what it comes up with. We can always adjust individual traces later.

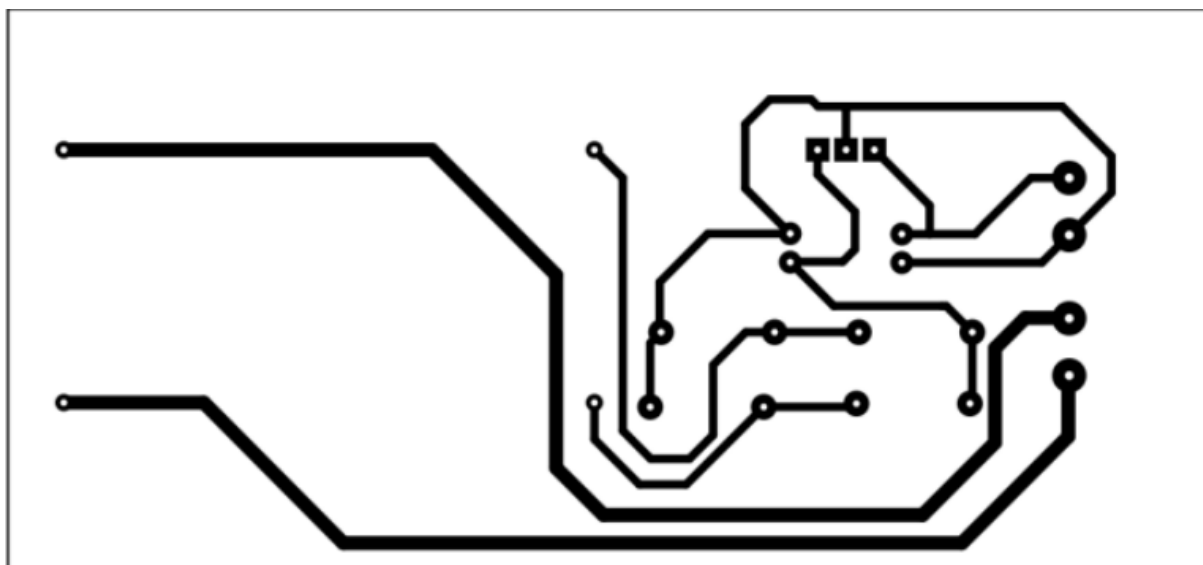
Now we have to Click on Output To get the Top Silk & Bottom Copper Layout print



PCB layout Printing options in Proteus



PCB layout (Top Silk)



PCB layout (Bottom Copper)

V. HARDWARE PREPARATION

i. Component Selection

1. Transformer with 1A 12V Rating
2. 4 pcs 1N4007 Diodes
3. 470uF & 220uF Electrolytic capacitor.
4. LDO or a Linear Voltage Regulator as per specification (Here LM7812 used).
5. A multi meter to measure the voltage.

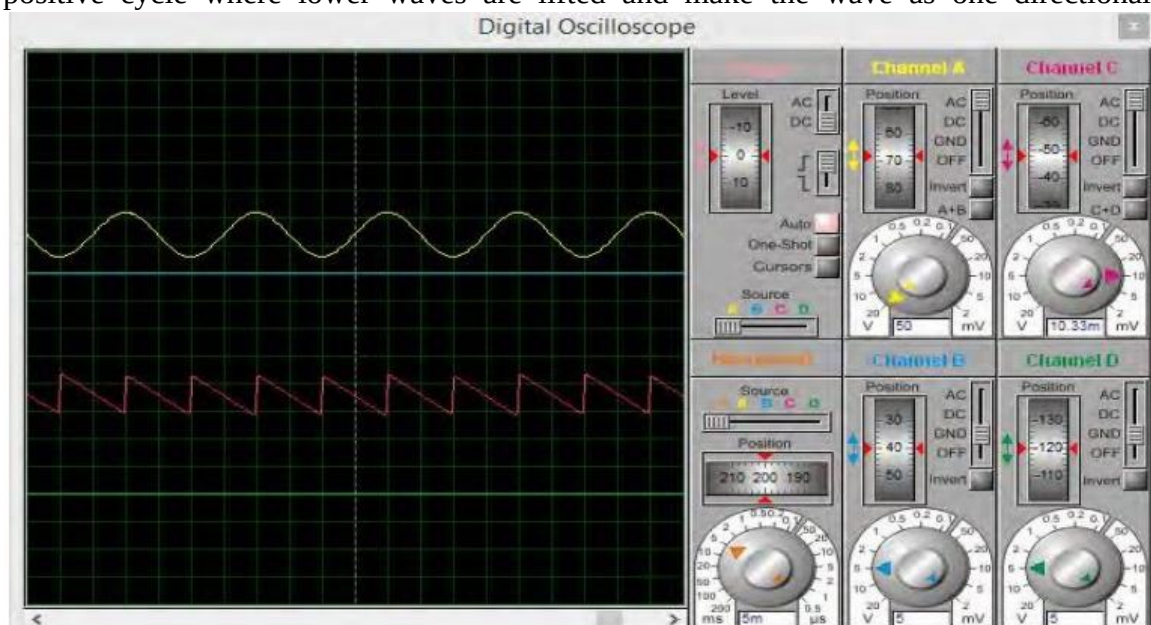
ii. Complete Hardware



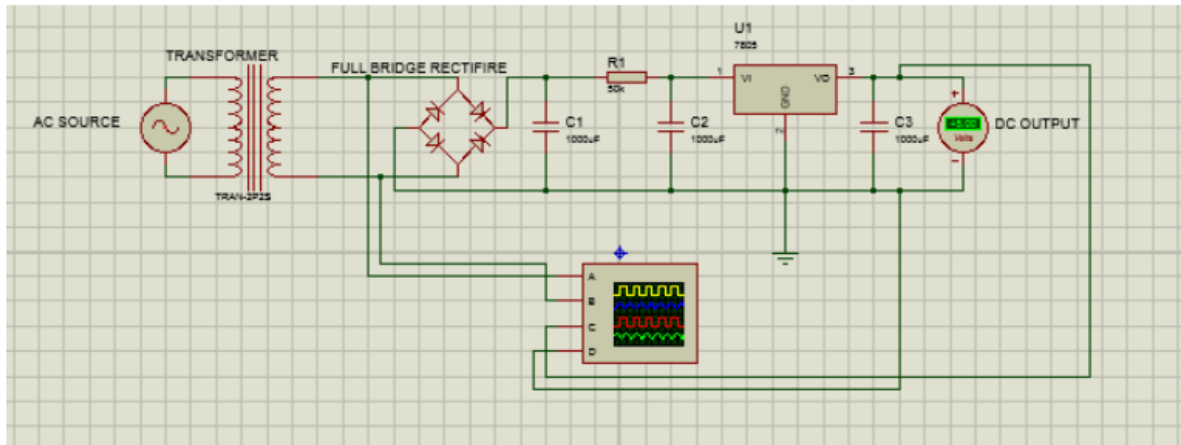
Complete Hardware Visuals (Ac to Dc Converter)

VI. RESULTS

From the circuit diagram shown in Fig 4, we give an AC source where AC voltage is given 220V and Frequency is 50Hz. Here, a transformer is attached with AC source which will transform 220v and make it a low voltage AC wave as AC voltage has positive (+Ve) and negative (-Ve) terminals. Here, after step down to a smaller voltage we need to stop polarity changes. To make electrons pass only in one direction we need diodes. Here we are using 4 diodes as a bridge so that a positive and negative cycle will convert into a positive cycle where lower waves are lifted and make the wave as one directional.



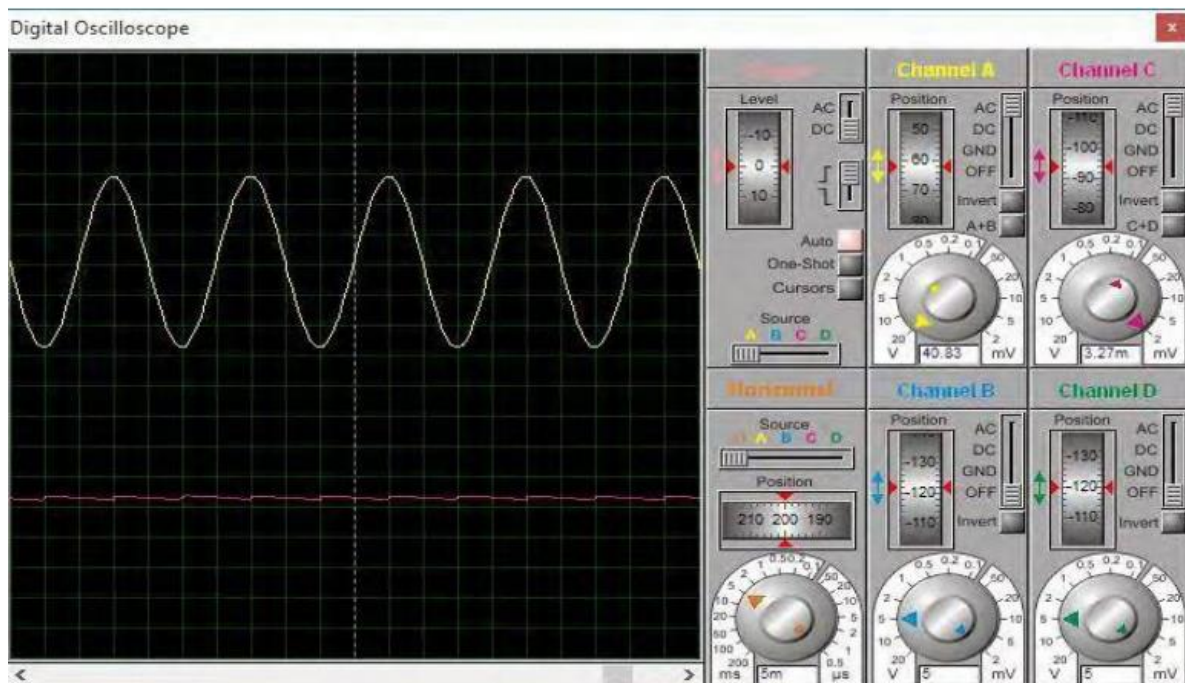
Simulation Graph of AC voltage as input with output of Full Bridge rectifier.



Circuit Diagram of AC voltage as input with output of DC voltage

Now, adding capacitor of higher value so that it can store charge. Here we are using Capacitors C1=470 μ F and C2=220 μ F. We can see that AC voltage come from the transformer as a sinusoidal wave, after that rectifier flips the negative half cycle into positive. Then with the help of capacitor the become smoother. Adding more capacitor, we can get smoother DC voltage. We can see the DC output which is almost straight with the help of capacitors.

From the circuit diagram, we can see the input from AC voltage coming from transformer and the output is from DC voltage source. A capacitor C3= 470 μ F is attached to store charge. Here we use oscilloscope for measuring the graph of AC input and DC output. We get output of DC voltage= +12V.



Simulation Graph of AC voltage as input with output of DC voltage.

Actually, line voltage is always presented. Filter capacitor fills those gaps as they don't get discharge very fast. It's smoothly generating DC. When the first DC wave comes up capacitor get charged and wave comes to zero instantly but capacitor discharge slowly, then

output line voltage is seen. As the capacitors don't get discharge very fast so capacitor fills those gaps which smoothly generates DC. Now if we want to make it straighter then we have to add more voltage regulator. This can be done by a zener diode also. Zener diode act as same as diode but it has a condition, limitation at a specific voltage. Here we are changing ac voltage to get our desired output. We get our dc voltage 11.95V when ac voltage is 220V.

NUMBER	AC VOLTAGE	FREQUENCY	C1 (μ F)	C2 μ F	C3 μ F	DC VOLTAGE
1	120V	50HZ	470	220	470	11.10V
2	150V	50HZ	470	220	470	11.13V
3	180V	50HZ	470	220	470	11.18V
4	220V	50HZ	470	220	470	11.95V
5	250V	50HZ	470	220	470	12.09V
6	280V	50HZ	470	220	470	13.02V

Practical result



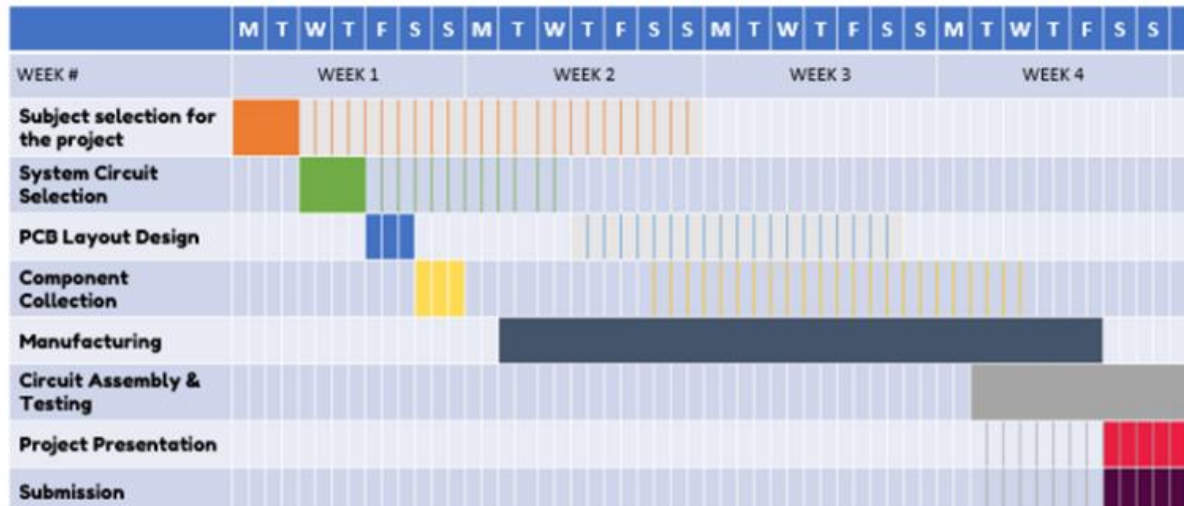
Hardware AC to Dc Converter Output Voltage

Here, we can see that we get almost our desired 12V output voltage. So, we can claim that our project is running properly.

Total Cost

[Estimated total cost associated with this project is around 600 Taka]

Timeline



VII. DISCUSSION AND CONCLUSION

AC to DC convert by using full bridge rectifier with different combination of load and observed the DC outcomes through simulations. We also studied about different types and application of AC-DC converters, keeping our main focus on single phase with full bridge rectifier converter circuits. We gained vast knowledge about PROTEUS software during the project, where we carried out all our circuit construction and simulation respectively.

The main function is to have a good conversion voltage from AC to DC, for that reason where we are using full bridge rectifier instead of normal diodes attaching or Zener diode. As we know Zener diode have limitation so when it reaches to that limitation it become bi-directional although it doesn't need voltage regulator but it will make bi-directional wave up to a limit. That is why we used full bridge rectifier and a voltage regulator to make a simple conversion. Besides, for better results we can use a voltage regulator chip or U0001f3a9 like 7805, 7812, 7905, 7912 to obtain a constant voltage (both positive and negative are possible). In Bangladesh the supply is 220Volts in 50Hz for households. We get 11.94V as DC output when we apply 220V as AC input.